

Rn (Rn I). Strongest classified transitions below 2 μm (NIST): 11 levels, 10 lines.

The diagram illustrates the energy levels and transitions for the $2p^2\ ^3P-^1D$ transition in Fe II . The energy levels are categorized by parity (even and odd) and total angular momentum quantum number (J).

Energy Levels (cm⁻¹):

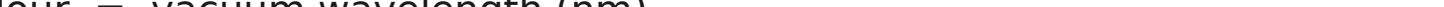
- Even Parity:**
 - $2[1/2]_0$: 69743.980
 - $2[3/2]_1$: 68332.100
 - $2[1/2]_1$: 66244.970
 - $2[3/2]_2$: 68789.930
 - $2[5/2]_2$: 66707.530
 - $2[5/2]_3$: 68039.480
 - $1S_0$: 0.000 (stable)
- Odd Parity:**
 - $2[1/2]_1^o$: 68891.340
 - $2[3/2]_1^o$: 55989.030
 - $2[3/2]_2^o$: 54620.350

Transitions (Wavelength in nm, Intensity):

- $2[1/2]_0 \rightarrow 2[3/2]_1$: 727.011 nm, 100
- $2[3/2]_1 \rightarrow 2[3/2]_2$: 810.171 nm, 100
- $2[1/2]_1 \rightarrow 2[3/2]_2$: 860.243 nm, 100
- $2[3/2]_2 \rightarrow 2[5/2]_2$: 781.195 nm, 100
- $2[5/2]_2 \rightarrow 2[5/2]_3$: 705.737 nm, 100
- $2[5/2]_3 \rightarrow 2[3/2]_1^o$: 745.205 nm, 100
- $2[3/2]_1^o \rightarrow 2[3/2]_2^o$: 178.606 nm, 100
- $2[3/2]_2^o \rightarrow 2[3/2]_1$: 827.323 nm, 100
- $2[3/2]_2^o \rightarrow 2[5/2]_2$: 827.323 nm, 100
- $2[3/2]_2^o \rightarrow 2[5/2]_3$: 827.323 nm, 100
- $2[3/2]_2^o \rightarrow 2[1/2]_1$: 860.243 nm, 100
- $2[3/2]_2^o \rightarrow 2[3/2]_1$: 810.171 nm, 100
- $2[3/2]_2^o \rightarrow 2[1/2]_0$: 727.011 nm, 100
- $2[3/2]_2^o \rightarrow 2[5/2]_2$: 781.195 nm, 100
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- $2[3/2]_2^o \rightarrow 2[5/2]_2$: 7

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e J' \rangle$ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
2[3/2] ₂ ^o - 2[5/2] ₃	745.2048	744.9996	402.2954	-	-	-	-	the line of the only laser-spectroscopy study of radon: c...

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)							
213/2 _{1/2}	-18.4(6)	2184(4)	F=3/2: +1335.81	F=5/2: +314.64	F=7/2: -568.91	F=9/2: -768.74	F=11/2: +417.30			
251/2 _{1/2}	-3.8(4)	2082(6)	F=1/2: +1166.25	F=3/2: +748.75	F=5/2: +258.75	F=7/2: -275.00	F=9/2: -604.25	F=11/2: -461.25	F=13/2: +481.25	

Measured A, B constants; sources in the data file.